The noise model, from zero signal processing

companion to *The residual bound, step by step*; equation numbers (19), (19b), (20) refer to that document

1 The problem in one sentence

The residual cap (20) needs one number per run: an upper bound on $\mathrm{RMS}(n)$, the root-mean-square of the disturbance n inside the fit window. The difficulty is that n is never observed alone — the gyro records $y = (\text{smooth rigid-body motion}) + n$, the two are stuck together in time, and there is no ground-truth curve to subtract. Everything in this note is one strategy for un-sticking them: *in time the two are inseparable, but in frequency they barely overlap.* The model motion is slow; much of the disturbance is fast. Split the record by frequency and the fast part is pure disturbance, measured, not assumed; the slow part is then recovered by a measured ratio. The note first builds the small signal-processing kit this needs (Sec. 2), then assembles the model (Secs. 3–6), and closes with why the obvious simpler alternatives fail (Sec. 7) and what the campaign checks show (Sec. 8). A glossary is at the end.

2 A five-minute signal-processing kit

2.1 A record is a vector

The window contains N samples $r_1, \ldots, r_N$ taken every T_s seconds (here $T_s \approx 5$ ms, $N \approx 80$–300, window length $T = NT_s \approx 0.4$–0.7 s). Treat them as one vector $r \in \mathbb{R}^N$. Its size is measured by the root mean square,

$$\mathrm{RMS}(r) \;=\; \sqrt{\tfrac{1}{N}\textstyle\sum_i r_i^2} \;=\; \frac{\|r\|}{\sqrt{N}}, \tag{N1}$$

which is just the vector's Euclidean length, normalised so it does not grow with N. Everything below is linear algebra on this vector; no probability is needed until the very end.

2.2 The DFT: the same vector in new coordinates

The discrete Fourier transform (DFT; the FFT is merely its fast implementation) does not modify the record. It re-expresses the same vector in a different orthogonal basis: instead of "sample 1, sample 2, …" the coordinates become "how much of a sinusoid at frequency f_k is in the record", for the specific grid of frequencies

$$f_k \;=\; \frac{k}{T}, \qquad k = 0, 1, 2, \ldots \tag{N2}$$

Each grid frequency is called a *bin*. Two things about (N2) are worth pausing on. First, the spacing is $\Delta f = 1/T$: a longer window gives finer frequency resolution — our windows give $\Delta f \approx 1.4$–2.5 Hz, so the shortest windows have only two or three bins below 5 Hz. Second, $k = 0$ is the *DC bin*, the record's mean; we always subtract the mean first, so it plays no role. For a real-valued record the negative-frequency half of the transform is the mirror image of the positive half, so one keeps only $k \geq 0$ (this is the `rfft`). The largest usable frequency is the Nyquist frequency $1/(2T_s) \approx 100$ Hz — faster oscillations cannot be represented by samples this far apart.

Each bin carries an amplitude: bin k's amplitude is the size of the sinusoid at f_k contained in the record. A bar chart of amplitude against f_k is the record's *spectrum* — Fig. 1(a) shows the actual bins of one run.

2.3 Parseval: Pythagoras in disguise

Because the sinusoids of the DFT are an *orthogonal* basis — perpendicular directions in $\mathbb{R}^N$, exactly like rotated coordinate axes — the squared length of the vector is the sum of squared coordinates in either basis:

$$\textstyle\sum_i r_i^2 \;=\; (\text{sum of squared bin contents}). \tag{N3}$$

This is Parseval's identity, and it is nothing deeper than Pythagoras: energy is counted once per perpendicular direction, whether the directions are "samples" or "frequencies". Its practical meaning: $\mathrm{RMS}(r)^2$ can be split *exactly* across groups of bins.

2.4 Bands, and the brick-wall split

A *band* is a set of bins — "below 5 Hz" (the low band), "above 5 Hz" (the high band). The *brick-wall split* at a cutoff f_c is the crudest possible filter and, for our purpose, the best one: transform the record, hand every bin *wholly* to one side of f_c, transform each side back. This produces two time records $r = r_{\mathrm{lo}} + r_{\mathrm{hi}}$ that are built from *disjoint* sets of basis directions, hence exactly orthogonal ($\sum_i r_{\mathrm{lo},i}\, r_{\mathrm{hi},i} = 0$), hence

$$\mathrm{RMS}(r)^2 \;=\; \mathrm{RMS}(r_{\mathrm{lo}})^2 + \mathrm{RMS}(r_{\mathrm{hi}})^2 \qquad \text{with no cross term, no approximation.} \tag{N4}$$

Checked on all 140 runs, the inner product and the Pythagoras error are 6×10^{-16} — machine precision. A "smoother" filter (a Butterworth-shaped mask, the kind used in real-time control loops) would share the transition-band bins *partially* between the two sides; the parts are then no longer orthogonal, and on the same data (N4) acquires a 19% error. The brick wall is what buys the equals sign, and the equals sign is what the whole construction leans on.

2.5 Where a smooth curve lives

The rigid-body motion in the window — the cosh branch, its error e_ω, the bias — evolves on the pendulum timescale: its spectral home is $C_2/2\pi \approx$ 0.6–0.9 Hz, and being a smooth exponential-type curve its bin amplitudes fall away steeply above a few hertz. Above 5 Hz the model family can produce *nothing*: whatever the record contains there is disturbance, by construction rather than by assumption. This single fact is the foundation of the model; Fig. 1(b) shows it directly — the fitted branch (orange) collapses above a few hertz while the record (blue) does not.

2.6 Leakage: the finite-window trap

One honest complication. The DFT silently treats the N samples as one period of a repeating signal. A record that ends at a different level than it began (ours rise monotonically) therefore contains an artificial jump at the seam, and a jump excites *every* bin — this is *(truncation) leakage*, and it is why the high band of the *raw* record is useless: the leaked tail of the smooth branch is 4–14× the disturbance up there. There are two standard cures, and the model uses both in different places:

- *Subtract before splitting.* If a curve carrying the same trend is subtracted first, its leakage is subtracted too; the remainder is nearly seam-free and its high band is clean. This is what both the residual anchor (Sec. 3) and the Savitzky–Golay anchor (Sec. 5) do — they differ only in *which* curve is subtracted.
- *Taper for display.* Multiplying the record by a window that fades to zero at both ends (a Hann taper) removes the seam and gives readable spectra. The display panels of Fig. 1(b) are tapered; the *numbers* never are — they come from the raw bins of panel (a), where subtraction has already cancelled the leakage.

2.7 Quadrature addition, and the shape ratio

Rearranging (N4): if the low/high ratio $\kappa = \mathrm{RMS}(r_{\mathrm{lo}})/\mathrm{RMS}(r_{\mathrm{hi}})$ is known, the total follows from the high band alone,

$$\mathrm{RMS}(r) \;=\; \mathrm{RMS}(r_{\mathrm{hi}})\sqrt{1+\kappa^2}. \tag{N5}$$

Orthogonal parts add like perpendicular legs of a triangle ("in quadrature"), never linearly. The factor $\sqrt{1+\kappa^2}$ converts a *level* (how big the high band is) using a *shape* (how the energy is distributed across frequency). The entire noise model is (N5) applied to the disturbance: measure the level where the model cannot reach, supply the shape from measurements the model cannot influence.

3 The anchor: the high band of the residual

The *minimiser residual* $r = y - \hat{\omega}$ (record minus fitted branch) has had the trend subtracted, so its high band is leakage-free; and above 5 Hz the subtracted branch changed nothing, since the model has nothing there.

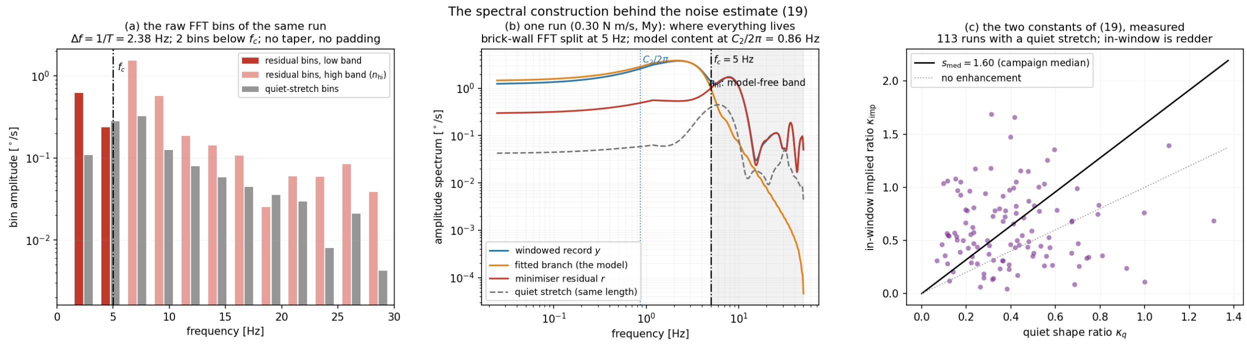

Figure 1: The construction, drawn. *(a)* The raw FFT bins of one run's residual and quiet stretch — one bar per bin $f_k = k/T$, split at $f_c = 5$ Hz, no taper: these are the exact objects every number below sums over. *(b)* Smoothed display spectra of the same run: the record (blue), the fitted branch (orange) collapsing above a few hertz, the residual (red) riding the record in the high band, the quiet stretch (grey) below it. *(c)* The two measured ratios across the campaign: quiet shape κ_q against in-window implied $\kappa_{\rm imp}$; the line is the median enhancement $s_{\rm med} = 1.60$.

Hence

$$n_{\rm hi} \;\equiv\; r_{\rm hi} \;=\; y_{\rm hi}, \qquad \mathrm{RMS}(n_{\rm hi}) \;=\; \text{the in-window disturbance level, measured.} \tag{N6}$$

$\mathrm{RMS}(n_{\rm hi})$ is the one number taken from the run on trust — no reference curve, no probability model; the separation is by frequency alone. One subtlety must be stated precisely: the anchor is not *fit-free* (subtracting *some* smooth curve was necessary, Sec. 2.6), but it is *fit-robust* — swapping the deployed fit for an entirely different member of the family changes $\mathrm{RMS}(n_{\rm hi})$ by 1.4% at the campaign median. The high band does not care which good smooth curve was removed; it only needed the seam gone.

4 From the high band to the full band: two measured ratios

(N5) now needs the disturbance's shape ratio κ. Two measurements deliver it, from opposite directions.

The quiet ratio κ_q. Before the excitation the vehicle sits on its gear with rotors at idle: no model motion at all, so a stretch of that record — cut to the *same length* as the window and split by the *same* brick wall — gives the disturbance's own low/high ratio with nothing to subtract. Campaign median $\kappa_q \approx 0.37$: quiet-time disturbance is mostly high-frequency.

The implied ratio $\kappa_{\rm imp}$. The window itself also implies a ratio, by inverting Pythagoras: (N4) gives $\mathrm{RMS}(r_{\rm lo}) = \sqrt{\mathrm{RMS}(r)^2 - \mathrm{RMS}(n_{\rm hi})^2}$, both quantities already in hand, so

$$\kappa_{\rm imp} \;=\; \frac{\sqrt{\mathrm{RMS}(r)^2 - \mathrm{RMS}(n_{\rm hi})^2}}{\mathrm{RMS}(n_{\rm hi})}. \tag{N7}$$

Its low band contains any unabsorbed model remainder as well, so $\kappa_{\rm imp}$ errs *high* — the safe direction: model leftovers get charged to the noise term, never hidden.

The enhancement $s_{\rm med}$. In the window the vehicle pivots on a gear edge under a growing moment; that contact pumps the *low* band, so the in-window disturbance is "redder" than quiet-time disturbance. The campaign median of $\kappa_{\rm imp}/\kappa_q$ over the 113 runs with a usable quiet stretch is $s_{\rm med} = 1.60$ (Fig. 1(c)) — in-window shape $\approx$ quiet shape scaled by 1.6.

5 The estimate (19), and the Savitzky–Golay anchor

The estimate. Assembling (N5) with the corrected shape gives the main document's equation (19), $\hat{n} = \mathrm{RMS}(n_{\rm hi})\sqrt{1 + (s_{\rm med}\kappa_q)^2}$ — the best available *estimate* of $\mathrm{RMS}(n)$, used for interpretation (e.g. Sec. 7 of the main document shows the measured residual sits statistically at $\hat{n}$). It is deliberately *not* what the cap uses: a median-calibrated estimate is exceeded by roughly half the runs, and its anchor comes from the minimiser residual — the very object the cap bounds. A guarantee should touch neither.

The Savitzky–Golay anchor. To free the anchor from the fit entirely, replace the subtracted curve (Sec. 2.6) by a generic smoother that has never heard of the cosh family. A Savitzky–Golay (SG) filter slides a short

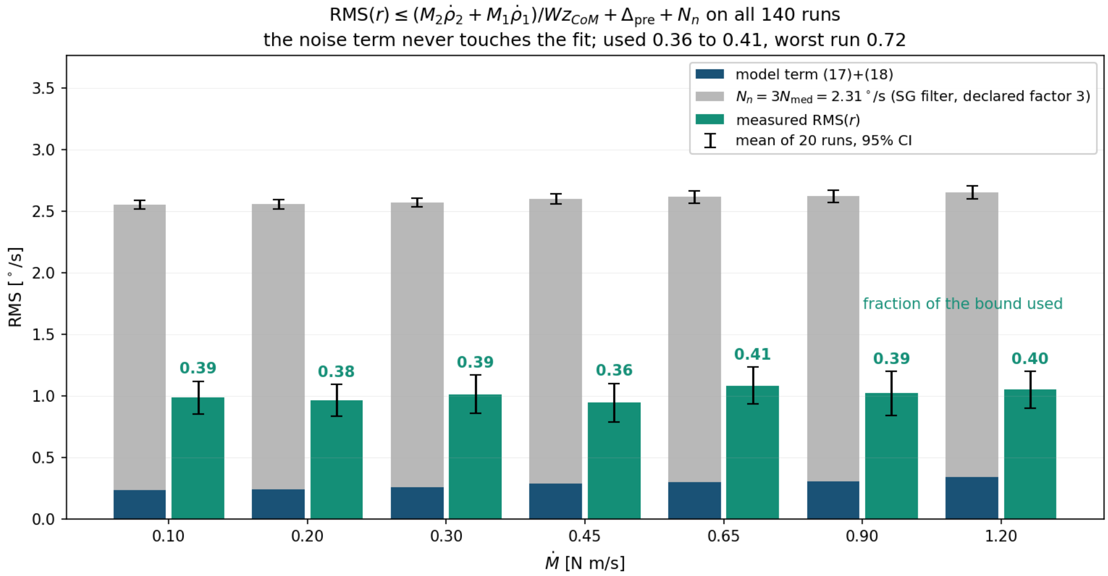

Figure 2: The cap (20) with the constant noise term (N9) against the campaign: model term (dark blue) + N_n (grey) versus the measured residual (green), means over 20 runs per rate with 95% intervals. All 140 runs inside; used fraction 0.36–0.41, worst run 0.72.

window along the record and replaces each sample by the value of a least-squares cubic fitted *locally* inside that window — a moving polynomial smoother, standard in chemistry and spectroscopy for a half-century. Its window length sets what it can follow: choosing $\approx 2/(f_c T_s)$ samples makes it track everything slower than about f_c and nothing faster, so (record − SG smooth) carries the disturbance's high band with the trend — and its leakage — removed. The same brick wall then reads off the level. Per run:

$$N_{\mathrm{SG}} \;=\; \mathrm{RMS}\big[(y - \mathrm{SG}(y))_{\mathrm{hi}}\big], \qquad N_{\mathrm{med}} \;=\; \mathrm{median}_{\mathrm{campaign}}\, N_{\mathrm{SG}} \;=\; 0.77^\circ/\mathrm{s}. \tag{N8}$$

Three cross-checks say this number is real, not an artefact of the smoother: a degree-4 Legendre polynomial detrend gives 1.00× the SG anchor at the campaign median, a DCT-based split gives 1.23×, and the fit-robust residual anchor of (N6) gives 1.03× — four independent routes, one level.

6 The adopted constant: $N_n = 3N_{\mathrm{med}}$

The cap's noise term is one campaign constant, the main document's (19b):

$$N_n \;=\; 3\, N_{\mathrm{med}} \;=\; 2.31^\circ/\mathrm{s} \qquad \text{for every run.} \tag{N9}$$

The factor 3 is *declared* — an engineering safety factor stated up front, not fitted — and it is sized by the two distinct jobs the constant must do at once:

job	factor	measured basis	acts
shape: high band → full band, (N5)	1.96	$\sqrt{1+\kappa^2}$ at the largest ratio 1.69	within-run
level: median anchor → every run	1.5	anchor max/median = 2.35	between-run

$1.96 \times 1.5 = 2.94$, rounded up to 3. The two halves cannot absorb each other: no shape factor, however large, covers a run whose *level* is twice the median. Empirically the minimum total factor that covers the campaign is 2.05, and the shape ceiling alone (1.96) leaves exactly the two level-outlier runs outside — the declared 3 carries 46% headroom over the minimum. The statement is falsifiable in advance: the noise term fails only if a future run's disturbance exceeds three times the campaign-median anchor. Fig. 2 shows the result: one grey block of identical height on every bar, no model curve, no minimiser, no per-run tuning anywhere in the noise side.

7 Why not the obvious simpler things

Each rejected alternative fails for a reason worth knowing, because each failure motivates one design choice above.

"Use the pre-onset noise floor times k." The quiet gyro floor is $\approx 0.17°$/s RMS — but the in-window disturbance is *contact-borne* (pivoting on a gear edge, motor load rising), not sensor noise, and runs 2 to 50× that floor, with no usable quiet stretch at all on 25 of 140 runs. One constant k covering the spread would be $k = 49$, collapsing the cap's used fraction to 0.16. *Lesson: the level must be measured in the window itself* — which is what (N6) and (N8) do.

"Use a textbook noise estimator." The two standard fit-free estimators — the second-difference formula and the finest-scale wavelet MAD — read 0.47× and 0.65× the true high-band level here. Both anchor at the very top of the band (near Nyquist), and this disturbance is *coloured* — its spectrum falls with frequency — so the top of the band is its quietest corner; the MAD variant additionally discounts bursts. Swapped into the cap they fail 59/140 and 13/140 runs. *Lesson: integrate the whole model-free band, not its quietest corner.*

"Split the raw record; skip the subtraction." Truncation leakage (Sec. 2.6) makes the raw high band read 4–14× too high. *Lesson: subtract a trend first; any good one works.*

"Keep the per-run envelope." The per-run form $\mathrm{RMS}(n_{\mathrm{hi}})\sqrt{1+\kappa_{\mathrm{sup}}^2}$ is tighter (mean used ≈ 0.50) but its anchor comes from the minimiser residual — defensible (it is fit-robust), yet a reader can still object that the method's own output appears in the method's own guarantee. The constant (N9) removes the objection at a known price.

"Take the campaign maximum; declare nothing." The maximum SG anchor with the shape ceiling gives 3.6°/s — every multiplier an observed extreme, no judgment anywhere — at mean used 0.26. Valid, and the natural endpoint for a maximally sceptical reader; (N9) sits one step tighter with one declared number.

8 What the checks show

On the 140-run campaign, with the model term of the main document: the cap holds on 140/140 runs; the used fraction is 0.36–0.41 by rate (per-run: p10 0.24, median 0.38, worst 0.72); the split's orthogonality is machine-exact (6×10^{-16}); moving the cutoff to 2 or 3 Hz changes nothing (140/140 either way) because a total does not depend on where it is split — only past 8 Hz, where too few bins remain above the cutoff, does the construction degrade; and the anchor is insensitive to the subtracted curve at the 1.4% level. The residual itself sits at the noise estimate ($\mathrm{RMS}(r)/\hat{n}$: median 0.976, IQR 0.89–1.09, flat across a twelvefold rate change) — the disturbance floor is what the fit leaves behind, and the model error underneath it (0.16–0.17°/s) is invisible at this floor, exactly as the theory predicts.

Glossary

bin	one frequency $f_k = k/T$ of the DFT grid; the record's content at that frequency
Δf	bin spacing $= 1/T$; longer window $\Rightarrow$ finer frequency grid
DC bin	the $k = 0$ bin = the record's mean; removed before everything
Nyquist	the fastest representable frequency, $1/(2T_s)$
spectrum	bin amplitude drawn against frequency
band	a set of bins ("below 5 Hz", "above 5 Hz")
brick-wall split	every bin handed wholly to one band; makes the two parts exactly orthogonal
orthogonal	perpendicular as vectors: $\sum_i a_i b_i = 0$; sizes then add by Pythagoras
Parseval	Pythagoras across the DFT basis: total energy = sum of bin energies
quadrature	adding orthogonal parts as $\sqrt{a^2+b^2}$, never $a+b$
leakage	spill-over of a trend into all bins, caused by the finite window's artificial seam
taper (Hann)	fading the record to zero at both ends to remove the seam; display only here
coloured noise	noise whose spectrum is not flat (here: falling with frequency)
SG filter	Savitzky–Golay: a sliding local least-squares polynomial smoother; a generic lowpass
quiet stretch	pre-excitation segment, rotors at idle: disturbance with no model motion in it
κ	a low-band/high-band RMS ratio: the *shape* of a spectrum in one number
anchor	the measured high-band level everything else scales from